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応用理工物理学実験
レポート

テーマ No.	IV 実 験 課 題	
IV	Polarization/Interference/Diffraction	
実験実施 月 日	第 1 回	第 2 回
	2025 月 06 日 23	2025 月 06 日 30
班	理工学群 応用理工学類 2 年 クラス N 番	
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Fundamental Labs II Report IV. Polarization/Interference/Diffraction

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Day 1(Week 1) Polarization

1.1 Purpose

To perform an experiment with polarized light to understand the nature of light as a transverse wave.

1.2 Principle

Light is a transverse wave. This means that the displacement of the particles of the medium during oscillation is perpendicular to the direction of propagation. Polarization refers to the orientation of the oscillations of light waves. Normally, light waves vibrate in many directions perpendicular to the direction of travel. A polarization filter is an optical filter that allows light waves of a specific polarization, that is, light oscillating in a single plane, to pass through while blocking waves of other polarizations.

1.3 Measurement of polarizing plate characteristics

1.3.1 Experimental Setup and Procedure

1) Preparation for Experiment

The laser beam of the experimental setup shown in Figure 1 is turned on. The polarizing plate is removed and the goniometer was rotated to 180º to make sure all the light enters the slit. The power meter is turned on and set to display the correct zero point.

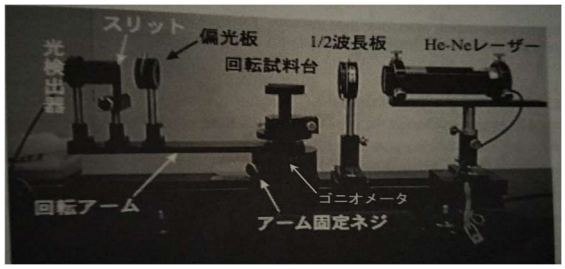


Figure 1

2) Measuring light intensity

After removing the metal plate blocking the laser light, we measure the intensity every 20 seconds 10 times until 200s to

confirm that the laser beam output is stable. The intensity is measured in mW.

3) **Measuring polarizing plate characteristics**

After attaching the polarizing plate to the rotating arm, it is adjusted so that it is perpendicular to the laser. While referring to the power meter, the rotation angle of the polarizing plate is set to lowest transmittance and this angle is recorded. The plate is slowly rotated from 0 to 90º and the power is recorded every 10º.

1.3.2 Discussion

1) **Measuring light intensity:**

The following table, Table 1, shows output as measured every 20 seconds 10 times until 200s.

Table 1 Measuring light intensity

times	time(s)	intensity(mW)
1	20	0.986
2	40	0.988
3	60	0.999
4	80	0.99
5	100	0.983
6	120	1.042
7	140	1.08
8	160	1.092
9	180	1.053
10	200	1.106

$I_0 = Average\ Intensity = 1.030692$ mW

We made a scatter plot in Excel to show the relationship between light intensity(mW) and time (s)

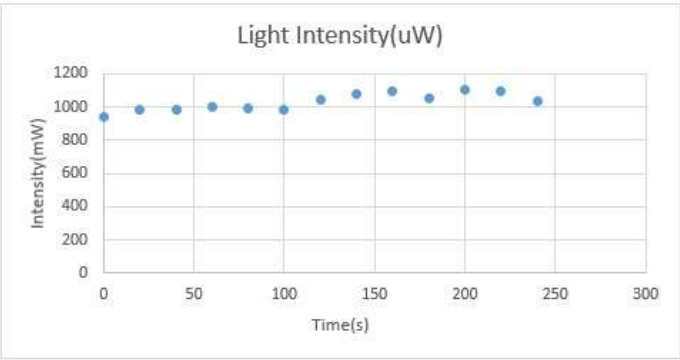


Figure 2 Light intensity (microW) vs time(s) until time(s)

2) Measuring polarizing plate characteristics

At the angle with the lowest transmittance, the direction of the electric field vector and the direction of polarization are orthogonal to each other (θ = 90). The lowest transmittance was observed to be at at α = 82.5 and since 90-82.5= 7.5, θ = α -7.5 The relationship between positive θ and the rotation angle was thus determined and tabulated below in Table 2.

Table 2 Measuring the I/I0 ratio with 〖cos〗^2 θ to find Ta and Tb with I0=Average Intensity= 1.030692 mW

α	θ(degree)	Intensity(microW)	θ(rad)	cos^2(θ)	I/I0
2.5	10	550	0.174533	0.96984631	533.6221
12.5	20	490	0.349066	0.883022222	475.4088
22.5	30	450	0.523599	0.75	436.5999
32.5	40	380	0.698132	0.586824089	368.6843
42.5	50	307	0.872665	0.413175911	297.8581
52.5	60	192	1.047198	0.25	186.2826
62.5	70	98	1.22173	0.116977778	95.08175
72.5	80	33	1.396263	0.03015369	32.01732
82.5	90	2.6	1.570796	3.7494E-33	2.522577

The graph showing the relationship between the ratio of the intensity of light T= I/I_0 and $cos^2\theta$ is shown Figure 3 below with $I_0 = Average\ Intensity = 1.030692$ mW as the previous experiment.

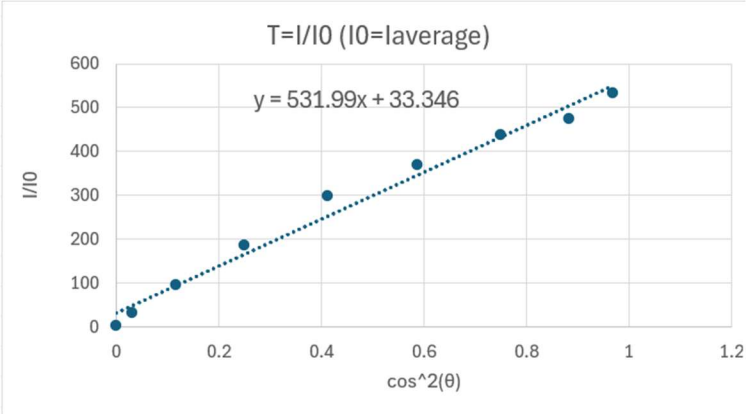


Figure 3 : I/I0 versus (cos θ)^2

From the figure, the graph shows a linear relationship y=mx+b.
From T= $T_a cos^2\theta + T_b sin^2\theta$, we can find T_a and T_b:
y=mx+b with m= slope and b= intercept
T= $T_a cos^2\theta + T_b sin^2\theta$= $T_a cos^2\theta$+$T_b - T_b cos^2\theta$
=$(T_a - T_b)\ cos^2\theta$+T_b
Slope of the graph m= $T_a - T_b$ and the intercept b=T_b
With the trendline= 531.99x+33.46, where y=T and x= $cos^2\theta$,
T_b= 33.346 and
T_a =m+b= $(T_a - T_b) + T_b$= 531.99+33.346=565.336

1.4 Measurement of Polarization Angle

1.4.1 Experimental Setup and Procedure

1. **Preparation for Experiment**

   As shown in the figure below the polarizing plate on the rotating arm is removed and the sample table is rotated to form an angle of 45º with the optical axis of the laser beam. A prism is placed on the step table to measure reflectance. The angle of the half-wave plate is set to 45º. The table is rotated to a point where the reflection intensity is the weakest and is adjusted to hit the photo-detector. At this point, the angle of the polarization plate is adjusted to output a minimum intensity. This angle (P-polarization angle) is recorded. The half-wave plate is now set to 0º and the above steps are repeated to obtain the S-polarization angle.

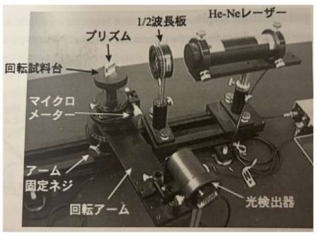


Figure 4

2. **Measuring polarization angle:**

   The intensity is measured every 10º for ±40º for the recorded arm angle using the power meter. More detailed measurements are made centering the arm angle where reflection intensity is the

smallest (a range of ±5º for every other degree). The above procedure is repeated for the recorded S-polarization angle.

1.4.2 Discussion

1. Measuring Polarization Angle

P Polarization:

The arm angle with the minimum intensity when the half-wave plate is set to 45º (P polarization) was 116º. The incident angle is calculated using arm angle/2

The results of the measured intensity are shown in the table below Table 3, with intensity measured in μW. The intensity results from the P is divided by $I_0 = Average\ Intensity = 1.030692$ mW.

Table 3 Finding minimum intensity with P polarization (Polarization angle)

arm angle (	angle of incidence(degrees)	intensityP	I/I0 (%) for
66	33	52	4.92145
76	38	49.5	4.68484
86	43	46.86	4.43498
96	48	38	3.59644
106	53	37.32	3.53208
111	55.5	37.98	3.59455
113	56.5	40.4	3.82359
115	57.5	39.35	3.72421
116	58	37.9	3.58698
117	58.5	41.2	3.8993
118	59	42.3	4.00341
120	60	43.74	4.13969
130	65	62.25	5.89154
140	70	100	9.46432
150	75	191.8	18.1526

S Polarization

The arm angle with the minimum intensity when the half-wave plate is set to 0º was 76 º. The incident angle is calculated using arm angle/2

The results of the measured intensity are shown in the table below, with intensity measured in μW.

The intensity results from S polarization is respectively divided by $I_0 = Average\ Intensity = 1.030692$ mW.

The results are in Table 4 below

***Table 4** Finding minimum intensity with S polarization (Polarization angle)*

arm angle (	angle of incidence(degrees)	intensity S	I/I0 (%) for S
66	33	119	11.2625
76	38	114.9	10.8745
86	43	151	14.2911
96	48	175	16.5626
106	53	204.5	19.3545
111	55.5	214.3	20.282
113	56.5	226.4	21.4272
115	57.5	232	21.9572
116	58	248.4	23.5094
117	58.5	250.8	23.7365
118	59	251.4	23.7933
120	60	271.5	25.6956
130	65	324.8	30.7401
140	70	407.5	38.5671
150	75	501.5	47.4636

I/I_0 for P (%) and I/I_0 for S (%)for were graphed against the incident angle to generate Figure 5. In the the reflectivity of S and P polarization , the ratio changed according to the incident angle.

Also in Figure 5, from the I/I_0 values for P Polarization and the incident angle graph, I rescaled the graph to show the Rp% graph for P Polarization in reference to the experiment textbook in Figure 6

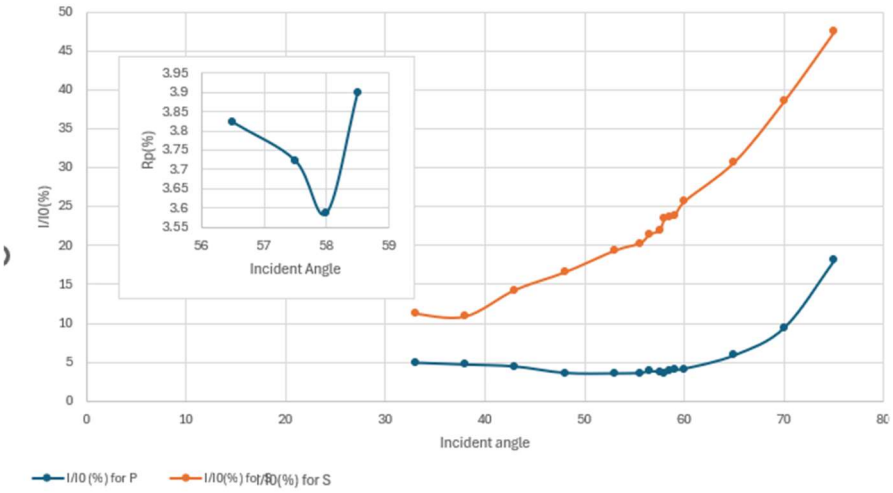


Figure 5 Intensity ratio (%) versus angle of incidence for S and P polarized light and Rp% versus angle of incidence

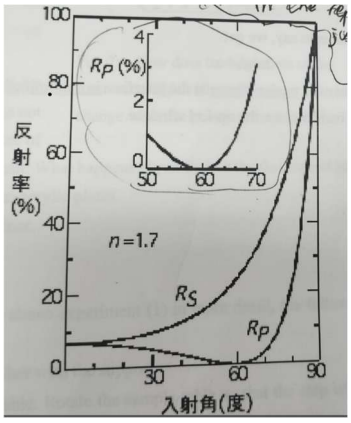


Figure 6 The change in reflectivity against the incident angle and Rp% in the textbook as reference

According to the tabular values and from looking at the trend on the graph, it was found that the minimum angle of incidence for P polarization was 116/2 = 58º.

The S polarization values follow the theoretical trend and the graph has a steeper curve and slope , almost following a linear graph than the P polarization value. The minimum value was 76 /2= 38 º.

Unlike in the textbook, the minimum value for the intensity ratio we obtained for our experiment did not reach zero.

2. Discussion Questions

(1) The difference between linearly polarized light and nonlinearly polarised light:

Linearly polarized light refers to light that has oscillations in a single plane and the electric field component only flows in one plane.

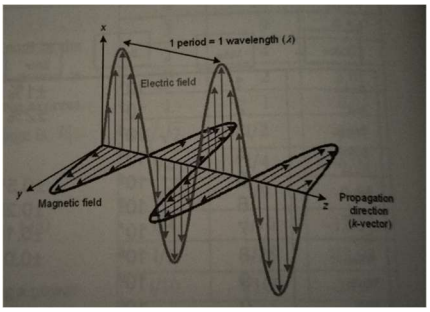


Figure 7 Linearly polarized light: Electromagnetic wave with linear polarization

Whereas non-linearly polarized light refers to light where electric field vector does not oscillate in a single straight line or plane.

Examples of non-linearly polarized light

-Circular polarized light

-Elliptical polarized light

(2) Explain the reason why brightness is changes when the polarizing plate was turned in the experiment where fluorescent light of a lamp was reflected onto an acrylic plate. 6 Polarization filter blocks certain components of incident light. At 0º, this is only one component (horizontal or vertical), therefore the intensity hitting the plate and our eyes is reduced by half. When this filter is turned, a component of the other half is also reduced. This causes less intensity of light to pass thorugh and this, its brightness is reduced.

(3) Based in (2) above, explain why its effective to wear “polarized sunglasses” in the sea or mountains.

At the sea, sunlight (with harmful UV rays) is directly reflected by the water, hitting our eyes. This is can be damaging to our eyes. Moreover, when light is reflected from the water, it is also slightly polarized, which causes us to not be able to see what is underwater (especially when fishing). Wearing polarizing glasses blocks this high intensity of light coming in from different directions and allows us to see better underwater from above the surface when fishing. At the mountains, due to a high altitude, once again the UV ray intensity is strong and can be damaging to out eyes. Therefore, polarizing glasses are useful at blocking certain frequencies of light, including UV which makes it less bright and more comfortable to see.

Day 2(Week 2) Interference/Diffraction

2.1 Diffraction by Double-Slit Task 1

2.1.1 Purpose

The purpose of this experiment is to observe and analyze diffraction phenomena by calculating the slit spacing from the interference fringes caused by the double slits of He-Ne laser and compare it with the value measured with an optical microscope.

The experimental setup is in Figure 8 and the experimental equipment is in Figure 9.

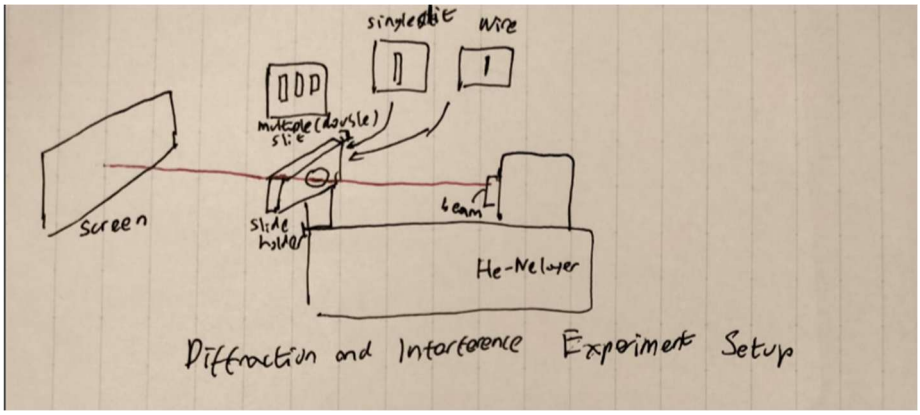


Figure 8 The diffraction and interference experiment setup in the experiment room

The experimental equipment is tape, paper, screen and the experimental equipment in the image below

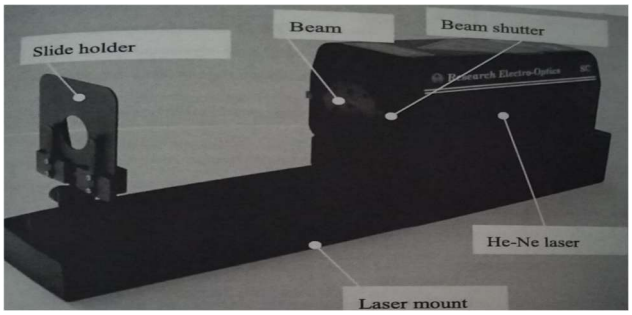


Figure 9 The experimental equipment. The slide holder can be changed to hold sample with double slit, single slit, or a wire.

2.1.2 Principle

Constructive interference in the double-slit experiment occurs when the path difference between the light waves from the two slits is an integer

multiple of the wavelength, λ. In double-slit interference, destructive interference only occurs at mλ/2 where m is an integer. And therefore, the condition dsinθ = mλ/2 is applicable.

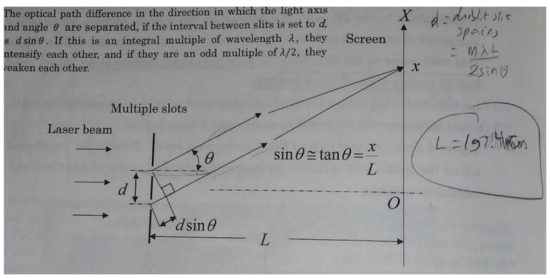


Figure 10

(Note : L is the distance between the slit and screen. In this figure, I marked our group's measured distance which is L=197.4cm)

2.1.3 Measurement using interference fringes observed from the laser light

1) Experimental method

(a) The He-Ne Laser light is passed through a double slit as shown in (figure) and is allowed to hit the screen with a white paper taped to it. The distance from the slit to the screen (L) is measured.

(b) The interference fringes are traced on the paper using a pencil. The distance from the center of the fringe in the middle to the center of the first destructive interference (X) is measured.

(c) This is repeated for the first three destructive interferences (or gaps where no light hit) as shown in Figure 11. The observations are tabulated.

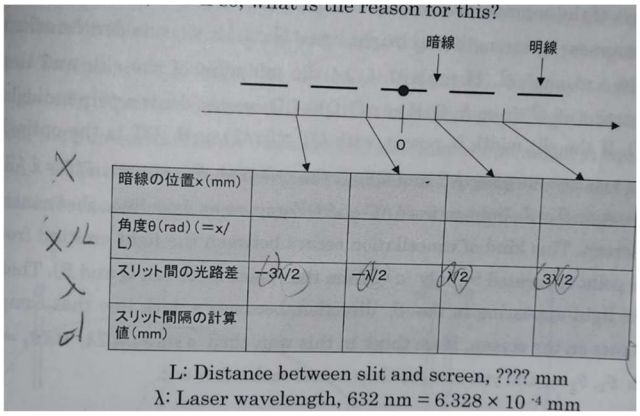


Figure 11

2) Results

The distance L was measured to be L=1974 mm and laser wavelength λ=6.328× $10^{-4}mm$

The slit spacing for an angular separation λ, d, was calculated in the following manner using the condition dsinθ = mλ/2 with sinθ=X/L, we obtained d=(mλL)/2X

where m is an integer, X is the distance between the mth destructive interference and O, and θ is the angle the ray maxes with the horizontal approximated by sinθ ≈ tanθ ≈ θ ≈ X/L. Table 5 shows the results of the calculated separation distance d for m =-3,-1, 1, 3.

For my own result, I noticed that for λ/2 and - λ/2, X and thus d is perfectly symmetrical by having the same values: X=3mm and d=0.208mm.

Table5 Double Slit Interference

		Double Slit Interference			
X(mm)	7	3	3	10	L
X/L(rad)	0.00355	0.00152	0.00152	0.00507	λ
λ	3λ/2	-1λ/2	1λ/2	3λ/2	
d(mm)	0.268	0.208	0.208	0.187	
d average(	0.218				

Since the value of d is different for every angular separation, albeit symmetrical on both sides, the average value can be calculated.

$$d_{average} = \frac{0.268+0.208+0.208+0.187}{4} = 0.218\text{mm}$$

2.1.4 Measurement of double slit spacing with an optical microscope

1) Experimental methods

The double slit was held under an optical microscope and the center line was aligned to the left edge of the first slit (a), the relative distance was noted. This is repeated for the right edge of the first slit (b), left edge of the second slit (c), and the right edge of the second slit (d) positions as in Figure 12. the separation distance (D) was calculated using the following formula.

$$D = \left| \frac{(a-b)}{2} + (b-c) + \frac{(c-d)}{2} \right|$$

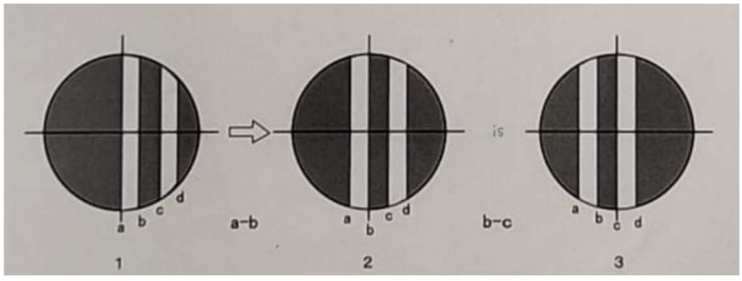


Figure 12 Method for measurement of double slit spacing with an optical microscope and markings for a,b,c, and d.

2) Result

We obtained a = 4.49mm, b=4.46mm, c= 4.77mm, and d= 4.94mm.

Thus D= 0.28mm

2.1.5 Observation

Unlike in previous experiments, this is a comparison between two different experiments, not comparing the vales obtained through experiment and that obtained by theoretical values.

Therefore, the differences in the value can be obtained by systematic error.

The error obtained in the double slit experiments (interference and microscope) is:
Error = $\Delta d = (0.28 - 0.218)/ 0.218 = 0.2844 \approx 28.44\%$

These differences in the value can be due to the following reasons:

• The angular separation is approximated by $\theta \approx \sin\theta \approx x/L$. As X increases, angular separation increases, making this approximation less accurate.
• The methodology has high scope of human error as we are marking the lines by hand using a pencil by tracing the interference pattern. Slightly longer or shorter lines by ±1 or 2 mm is highly possible, causing out answers to potentially change.
• Moreover, a ruler was used for measurements, making them less accurate due to measurement errors.
• The samples with the double slit, upon viewing from a microscope, had specks of dust and moisture on it. When the interference patterns were observed, the fringes were not clear (their ends were a little blurry and not defined). There was also scattering of light in other places due to the impurities that had to be ignored.
• For the microscope, error can be made by mistaken reading by a few millimetres of the microscope micrometre calibration ruler.

2.2 Diffraction by Single-Slit Task 2

2.2.1 Purpose

The purpose of this experiment is to observe and analyze diffraction phenomena by calculating the slit width from the interference fringes caused by the single slit of He-Ne laser and compare it with the value measured with an optical microscope.

2.2.2 Principle

1) Huygen's principle: Why there are patterns even when there's only a single slit

According to Huygens's principle, every point across the slit acts as a source of tiny wavelets. These wavelets spread out and interfere with each other.

- When light travels straight ahead, all wavelets from the slit travel the same distance to the screen, so they stay in phase and create a bright central maximum (constructive interference).
- When light travels at an angle, the wavelets travel different distances to the screen. For some angles, this causes them to arrive out of phase—for example, if the top ray travels one full wavelength more than the bottom ray, then each ray has a matching ray that cancels it out, creating a dark minimum (destructive interference).
- At other angles, the path difference may allow most wavelets to add up in phase, producing a bright fringe, but not as bright as the central one.
- The condition for destructive interference is that the path difference between the top and bottom rays equals an integer multiple of the wavelength (e.g., $d\sin\theta = m\lambda$, where $m = \pm 1, \pm 2, \ldots$)

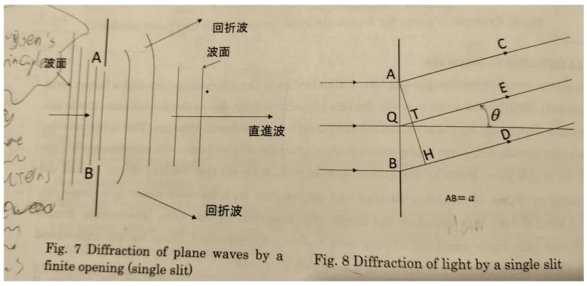


Fig. 7 Diffraction of plane waves by a finite opening (single slit)

Fig. 8 Diffraction of light by a single slit

Figure 13

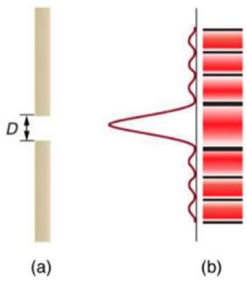


Figure 14 Single slit diffraction pattern

2) Constructive interference in the single-slit experiment occurs when the path difference between the light waves from the two slits is an integer multiple of the wavelength shifted by half, λ. Therefore, in single-slit interference, destructive interference only occurs at mλ where m is an integer. And therefore, the condition dsinθ = mλ is applicable.

2.2.3 Measurement using interference fringes observed from the laser light

1. Experimental Method:

(a) The He-Ne Laser light is passed through a single slit and is allowed to hit the screen with a white paper taped to it. (The thinner slit was used in this experiment)

(b) The interference fringes are traced on the paper using a pencil. The distance from the center of the fringe in the middle to the center of the first destructive interference (X) is measured.

(c) This is repeated for the first two destructive interferences (or gaps where no light hit). The observations are tabulated

2. Results:

The distance L was measured to be L=1974 mm and laser wavelength λ=6.328× $10^{-4}mm$

The slit spacing for an angular separation λ, d, was calculated in the following manner using the condition dsinθ = mλ with sinθ=X/L, we obtained d=(mλL)/X

where m is an integer, X is the distance between the mth destructive interference and O, and θ is the angle the ray maxes with the horizontal approximated by sinθ ≈ tanθ ≈ θ ≈ X/L. Table 6 shows the results of the calculated separation distance d for m =-3,-1, 1, 3.

Table 6 Single Slit Interference

		Single Slit Interference		
X(mm)	40	21	18	40
X/L(rad)	0.0203	0.0106	0.00912	0.0203
λ	-3λ	-1λ	1λ	3λ
d(mm)	0.0937	0.0595	0.0694	0.0937
d average(	0.0791			

For my own result, I noticed that for 3λ/2 and - 3λ/2, X and thus d is perfectly symmetrical by having the same values: X=40mm and d=0.0937 mm.

Since the value of d is different for every angular separation, albeit symmetrical on both sides, the average value can be calculated.

$$d_{average} = \frac{0.0937+0.0595+\ .0694+0.0937}{4} = 0.0791\text{mm}$$

2.2.4 Measurement of single slit spacing with an optical microscope

1. Experimental Method:

The experimental method is like the experimental method for measurement of double slit spacing in 2.1.4 and Figure 13, except that now we only measured a and b since there is only a single slit.

The single slit was held under an optical microscope and the center line was aligned to the left edge of the slit (a), the relative distance was noted. This is repeated for the right edge (b) and the slit width (D) was calculated using the following formula. As in Figure 15.

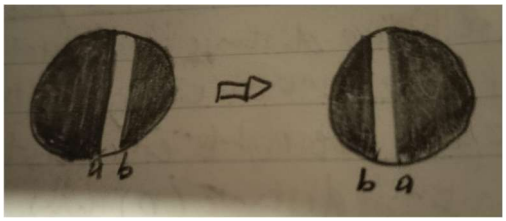


Figure 15 Measurement of single slit spacing with optical microscope. We now only have a and b to measure.

D= a-b

2. Results:

We obtained a= 10.08 mm and b=10.03 mm

D= a-b=0.05 mm

2.2.5 Observations

For the interference results, $d_{average}$ for single slit is roughly ½ of $d_{average}$ for double slit. This may have something to be with the conditions with the single slit and for the double slit: for the double slit dsinθ = mλ/2

with sinθ= x/L so d= mλL/2X and for the single slit dsinθ= mλ with sinθ= x/L so d= mλL/X.

However, mathematically the ratio should've been like this from the interference formulas:

Thus $\frac{d_{single\ slit}}{d_{double\ slit}} = \frac{1}{\frac{1}{2}} = 2$

The results possibly show an inverse relationship to this ratio in the experiment.

Because of similar reasons to the double-slit experiment, the d value is different for every separation distance in the first methodology.

The differences in the value of d can be obtained by systematic error.

The error obtained in the single slit experiments (interference and microscope) is:

Error = $\Delta d = (0.0791 - 0.05)/ 0.0791 = 0.367 \approx 36.7\%$

2.3 Diffraction by a Wire Task 3

2.3.1 Purpose

The purpose of this experiment is to observe and analyze diffraction phenomena by calculating the wire diameter from the interference fringes caused by the He-Ne laser and compare it with the value measured with an optical microscope.

2.3.2 Principle

Babinet's Principle states that the diffraction pattern produced by an opaque object is identical to that produced by a complementary aperture of the same size and shape, except for an overall difference in intensity. Therefore, the wire width can be calculated used a similar principle to single slit interference.

2.3.3 Measurement using interference fringes observed from the laser light
1. Experimental Method

2. Results:

The distance L was measured to be L=1974 mm and laser wavelength $\lambda=6.328\times 10^{-4}$mm

The slit spacing for an angular separation λ, d, was calculated in the following manner using the condition $d\sin\theta = m\lambda$ with $\sin\theta=X/L$, we obtained $d=(m\lambda L)/X$

where m is an integer, X is the distance between the mth destructive interference and 0, and θ is the angle the ray maxes with the horizontal approximated by $\sin\theta \approx \tan\theta \approx \theta \approx X/L$.

Table 7 Wire interference

Table 7 using Excel shows the results of the calculated separation distance d for m =-3,-1, 1, 3.

		Wire Interference		
X(mm)	16	5	5	17
X/L(rad)	0.00811	0.00253	0.00253	0.00861
λ	-3λ	-1λ	1λ	3λ
d(mm)	0.234	0.2498	0.2498	0.220
d average	0.239	mm		

For λ/2 and - λ/2, X and thus d is perfectly symmetrical by having the same values. For 3 λ/2 and -3 λ/2, X and thus d are almost perfectly symmetrical also with X=16mm and X=17mm, respectively.

Thus $d_{average} = \frac{0.234+0.2498+\ .2498+0.220}{4} = 0.239$mm

2.3.4 Measurement of wire width with an optical microscope

1. Experimental Method:

Since measurement of wire width is the same as the measurement of single slit using Babinet's principle, we measured the wire width with an optical microscope using the same experiment method as in 2.2.4 for measurement of single slit spacing with an optical microscope

D=a-b

2. Results:

a= 0.83 mm and b= 0.60mm

D= a-b = 0.23mm

2.3.5 Observations

The results obtained by using interference and by using optical microscope are close in value. This may be due to more accurate drawing of the interference fringes and more accurate measurement with the ruler.

The error obtained is only:

Error = $\Delta d = (0.239 - 0.23)/ 0.239 = 0.037 \approx 3.7\%$

Even though the wire measurement method is similar to the single slit measurement method, the values obtained by the wire measurement method are more similar to that obtained by the double slit measurement method.